- # Task A -Data Quality & Preprocessing:

1) Identify missing values, duplicates, and noisy/outlier data:

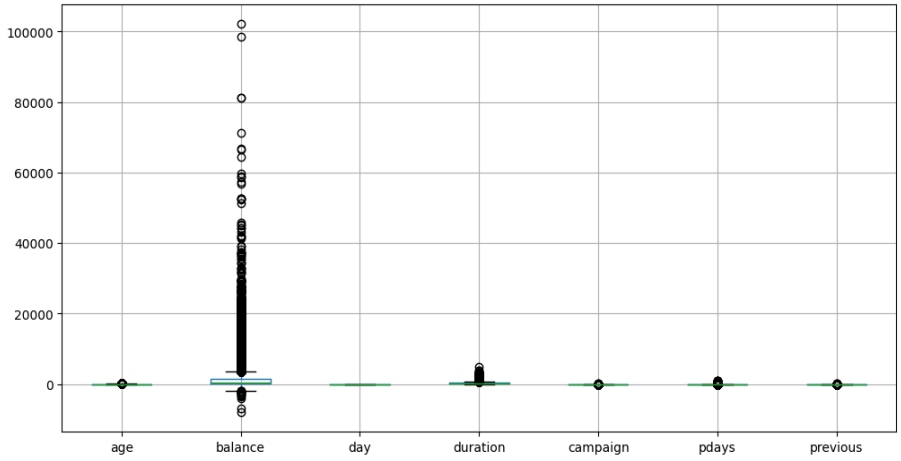
```
#   Column      Non-Null Count  Dtype
---  ------      --------------  -----
 0   age         45211 non-null  int64
 1   job         45211 non-null  str
 2   marital     45211 non-null  str
 3   education   45211 non-null  str
 4   default     45211 non-null  str
 5   balance     45211 non-null  int64
 6   housing     45211 non-null  str
 7   loan        45211 non-null  str
 8   contact     45211 non-null  str
 9   day         45211 non-null  int64
 10  month       45211 non-null  str
 11  duration    45211 non-null  int64
 12  campaign    45211 non-null  int64
 13  pdays       45211 non-null  int64
 14  previous    45211 non-null  int64
 15  poutcome    45211 non-null  str
 16  y           45211 non-null  str
```

Duplicates datapoints:

```
df.duplicated().sum()
```
✓ 0.3s

```
np.int64(0)
```

noisy/outlier data:



2) **Explain the types of attributes:**

```
Numerical features: Index(['age', 'balance', 'day', 'duration', 'campaign', 'pdays', 'previous'], dtype='str')
Categorical features: Index(['job', 'marital', 'education', 'default', 'housing', 'loan', 'contact',
       'month', 'poutcome'],
      dtype='str')
```

3) **Apply preprocessing:**
I have used a power transformer in which I apply OneHotEncoding on categorical features, and a standard scaler on numerical features.
I applied a Label encoder separately on the target column.

4) **Split the dataset into Train/Test:**

```
x_train, x_test, y_train, y_test = train_test_split(df.drop("y", axis = 1), df['y'], test_size = 0.2, random_state = 42)
✓ 0.1s
```

- ## Task B -Model Training + Evaluation:

1) **Logistic Regression (Baseline model):**
Classification matrix –

	precision	recall	f1-score	support
0	0.92	0.98	0.94	7952
1	0.65	0.34	0.45	1091
accuracy			0.90	9043
macro avg	0.78	0.66	0.70	9043
weighted avg	0.88	0.90	0.88	9043

- ## Task C -Decision Tree Learning + Interpretation:

Baseline model (without tuning)--

	precision	recall	f1-score	support
0	0.93	0.93	0.93	7952
1	0.48	0.49	0.48	1091
accuracy			0.87	9043
macro avg	0.70	0.71	0.71	9043
weighted avg	0.88	0.87	0.87	9043

With tuning –

	precision	recall	f1-score	support
0	0.91	0.97	0.94	7952
1	0.65	0.33	0.44	1091
accuracy			0.90	9043
macro avg	0.78	0.65	0.69	9043
weighted avg	0.88	0.90	0.88	9043

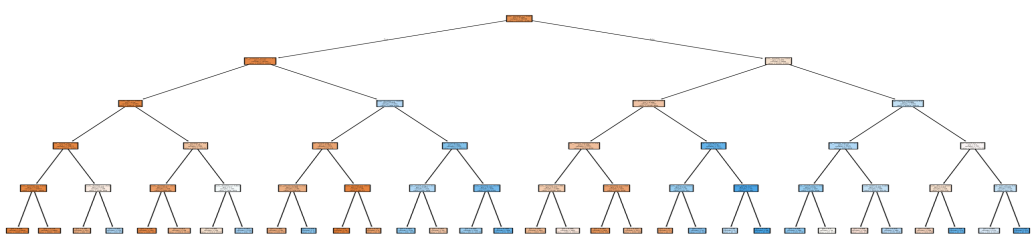
Both precision and accuracy have improved by tuning, but recall has reduced from 49 to 33.

Feature importance –
Features that have a greater say in prediction

feature	importance
num__duration	0.569681
cat__poutcome_success	0.299584
num__age	0.049397
cat__month_mar	0.039760
cat__contact_unknown	0.017580
cat__month_may	0.006090
cat__housing_no	0.004612
num__balance	0.002265
num__day	0.002221
cat__marital_married	0.001792

- ## Task D -Rule-Based Classification:

  Rule-Based Classification using a rule extraction approach from a decision tree



The decision tree is trained with max depth = 5, min samples split = 10, and min samples leaf = 5.

5 rules extracted from the decision tree —

```
1) Rule 1:
if duration <= 0.99 and poutcome_success <= 0.50 and age <= 1.84 then customer will not subscribe term deposit
2) Rule 2:
if duration <= 0.99 and poutcome_success <= 0.50 and age <= 1.84 and month = march and duraion > -0.40 then customer will subscribe the term deposit
3) Rule 3:
if duration <= 0.99 and poutcome_success >  0.50 and age > 1.84 and duration <= -0.46 and month = may then customer will not subscribe
4) Rule 4:
duration >  0.99 and duration <= 2.25 and poutcome_success <= 0.50 then customer will not subscribe
5) Rule 5:
duration >  0.99 and duration >  2.25 and contact_unknown <= 0.50 and marital_status = unmarried and month_nov <= 0.50 then customer will subscribe the term deposit
```

Explain how rules affect interpretability vs performance –

```
1) Interpretability:
(a) Rules are easy to interpret
(b) They can be used by businesses easily without the help of experts
2) Performance:
(a) Rules are generally used for simplification and are less capable of learning complex patterns.
```

- ## Task E -kNN (Lazy Learning):

```
==================================================
value of k= 1
accuracy score: 0.8814552692690479
==================================================
value of k= 3
accuracy score: 0.8925135463894726
```

```
==================================================
value of k= 5
accuracy score: 0.8984850160345018
==================================================
value of k= 7
accuracy score: 0.8979321021784806
==================================================
value of k= 9
accuracy score: 0.8995908437465443
==================================================
```

```python
metrics = ['euclidean', 'manhattan']

for (variable) knn: KNeighborsClassifier
    knn = KNeighborsClassifier(n_neighbors=5, metric=m)
    knn.fit(x_train_encoded, y_train_encoded)

    y_pred = knn.predict(x_test_encoded)

    print(f"\nMetric = {m}")
    print("Accuracy:", accuracy_score(y_test_encoded, y_pred))
```

✓ 25.1s

```
Metric = euclidean
Accuracy: 0.8984850160345018

Metric = manhattan
Accuracy: 0.8991485126617274
```

Why scaling is critical for kNN?

knn is distance based machine learning model. Whenever we give test data to the model, it first computes the distance with all datapoints then it identify k nearest neighbour as per the distance, and predict value by counting average of selected k values. In this process, distance becomes crucial component of prediction. To calculate the distance we use Euclidean and manhattan distance formulas, and if we give either of the values in a higher range, there are very high chances that the higher value dominates the result, suppressing the significance of the other value. This skewedness can cause incorrect prediction.

How does k affect bias/variance?

Low value of k: It led to high variance and low bias. It happens because the model finds the precise value of the train datapoint.
High value of k: It led to low variance and high bias. In such cases, the model does not find the precise value to predict for the

```
test data, rather it calculates the average of k nearest neighbor
datapoints.
```

- **Task F - Ensemble Learning:**

For random forest –

```
Random Forest Accuracy: 0.9034612407386929
              precision    recall  f1-score   support

           0       0.92      0.97      0.95      7952
           1       0.66      0.41      0.51      1091

    accuracy                           0.90      9043
   macro avg       0.79      0.69      0.73      9043
weighted avg       0.89      0.90      0.89      9043
```

For adaboosting –

```
AdaBoost Accuracy: 0.8938405396439235
              precision    recall  f1-score   support

           0       0.92      0.97      0.95      7952
           1       0.66      0.42      0.51      1091

    accuracy                           0.90      9043
   macro avg       0.79      0.69      0.73      9043
weighted avg       0.89      0.90      0.89      9043
```

- **Task G -Final Comparison + Recommendation:**

	Model	Accuracy	F1 Score	Notes
0	Logistic Regression	0.898706	0.448857	Baseline model
1	Decision Tree	0.873272	0.483318	prone to overfitting
2	Tuned Decision Tree	0.897490	0.439201	prone to overfitting
3	kNN	0.899591	0.449029	Sensitive to k
4	Random Forest	0.903461	0.507614	performance
5	Gradient Boosting	0.893841	0.413203	accuracy